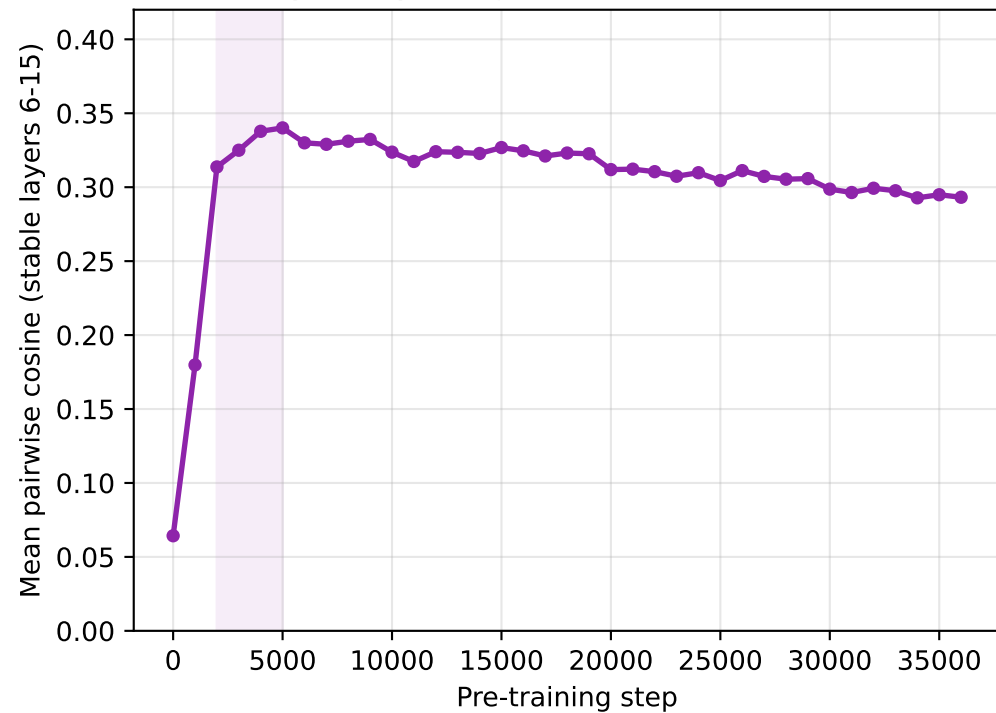
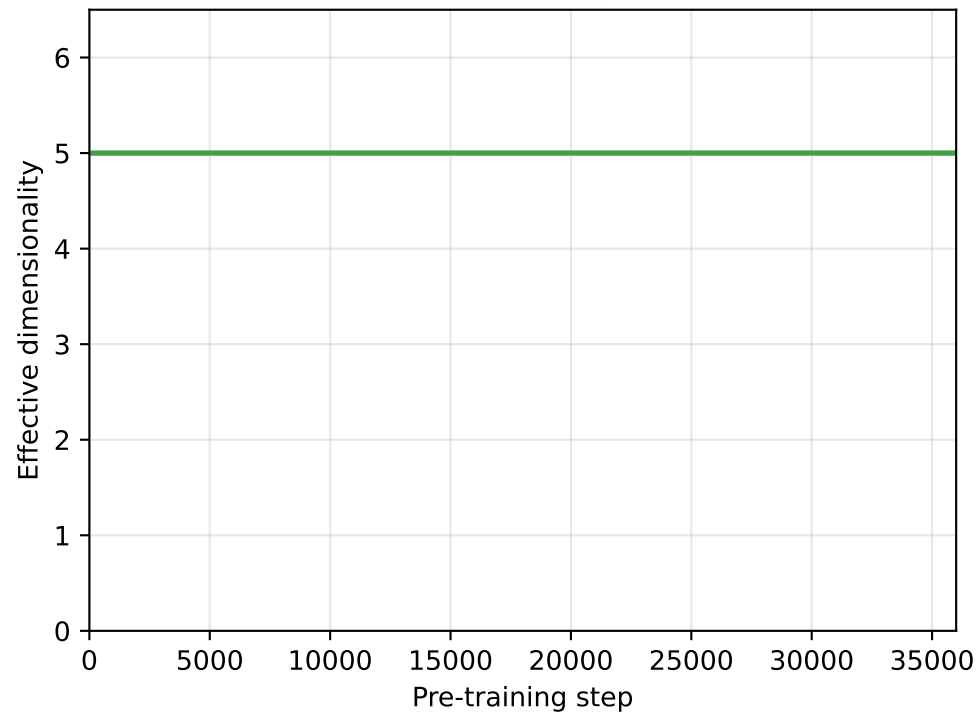


Geometric trajectory during OLMo-2 1B pre-training (37 checkpoints, seed-fixed directions)

(a) Geometry: early integration, then slow differentiation



(b) Effective dimensionality = 5 throughout



(c) Accuracy keeps climbing after geometry sets

